A Comprehensive Mobile Platform for Fostering Communication, Literacy, Numeracy, and Emotion Understanding in Children with ASD

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***Abstract*— This study presents SIPNENA, a novel mobile application designed to aid the learning and communication development of Sinhala-speaking autistic children aged six, particularly in rural areas of Sri Lanka. It offers a unique approach to teaching challenging subjects like English and Mathematics, tailored to the specific needs of children with Autism Spectrum Disorder (ASD). The application integrates interactive methodologies and gamification elements to facilitate better communication, understanding, and engagement. Additionally, it incorporates real-time emotion recognition features to monitor and respond to children's emotional states during learning activities. This research evaluates SIPNENA's effectiveness in improving communication abilities, academic skills, and emotion understanding among autistic children. The findings indicate promising results in catering to the unique educational needs of this target population, particularly in under-resourced rural regions, where specialized interventions are often scarce.**

Keywords—SIPNENA, Autism Spectrum Disorder, Mobile Learning, Rural education, Child development, Educational Technology

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a neurological condition characterized by social interaction, communication, and repetitive activities. Individuals with ASD face educational challenges, especially in rural areas [1], where specialized services and interventions are often scarce. The research on 'SIPNENA' aims to address these issues by focusing on communication abilities, which are often challenging for children with ASD. This research aims to provide tailored care and support for these individuals. The linguistic and cultural divide in Sri Lanka hinders the provision of appropriate assistance for individuals with ASD, as educational and technical resources are primarily available in internationally dominant languages [2], causing a divide for Sinhala-speaking autistic children. The study aims to address this issue by introducing a mobile application. 'SIPNENA' is a curriculum-based learning approach for autistic children, aiming to empower them to engage with complex subjects like English and mathematics [3], using a culturally and linguistically familiar medium, which is expected to enhance their learning experience and academic progress. Communication difficulties and challenges in understanding emotions are common among children with ASD. Several research papers have discussed communication interventions [3], but 'SIPNENA' goes beyond that by incorporating activities designed to improve communication skills and develop emotional understanding. These aspects are very important as they address the fundamental difficulties faced by individuals with ASD. The use of e-learning applications for autistic children in Sri Lanka is hindered by complexities in user interface design, leading to stress and difficulties in their use. These tools often have unfriendly interfaces, bugs, and steep learning curves. Parents and caregivers may lack the knowledge and abilities to effectively use these applications, which are often not reliable or user-friendly for ASD children. Despite the effectiveness of e-learning programs, the lack of use for this specific user group remains a significant issue. [4]

II. RELATED WORK

The development of effective learning and communication systems for ASD students is crucial to enhancing their educational experiences and facilitating effective communication. Despite numerous research studies, challenges persist, and there is a need for continued exploration and innovation in this domain. Each research endeavor has its own unique focus, and while they have significantly contributed to our understanding, there is still room for improvement and holistic solutions. These studies involve extensive reviews of existing research papers related to learning and communication systems for ASD students, examining the progress made and achievements attained by researchers in this critical area.

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The paper [6], explains a system that includes a Research based, graded Vocabulary, Symbols and Images, Measure the child's progress with Avaz analytics, presented as easy to understand graphs, Seamless Integrated Text and Keyboard Mode, and zoom and animation. is the first Augmentative and Alternative Communication (AAC) app launched in Sinhalese [5]. Zoom and animation features are provided using reinforcement learning. Word prediction is done using images and text. Children can say words and create sentences by touching a picture symbol that can speak with a voice. For example, if the child is hungry and cannot say the word "biscuit", he/she can touch the picture of a biscuit in the Avaz SL app and it will speak the word aloud. This app targets a dearth in resources of this nature in Sri Lanka especially in the local languages; there is no better opportunity that we can provide children who have difficulty speaking and their families than the ability to communicate better so that they too may have voice. The product fulfills several new features but does not fulfill several essential requirements such as recognizing emotions, maintaining a conversation and creating activities targeting the school curriculum.

Duolingo is a 2021 research study [7] that focuses on a language application that offers practical and systematic steps for learning a foreign language. The app, which teaches English to millions of people for free, differs from other free language learning apps by teaching through translation. This method, which was abandoned in the late 1960s due to the rise of communicative approaches, is based on web and mobile application technology, speech recognition for pronunciation training, and cloud-based infrastructure for scalability and personalization. Duolingo also uses text analysis, translation, speech-to-text, and natural language processing (NLP) for personalized learning paths. However, the application does not cover basic aspects such as identifying emotions, developing logical knowledge like mathematics, and improving communication skills.

In the year 2017, the paper [8] describes about 'AutiAid' mobile application which is done for ASD students. This provides a range of features, including interactive games, videos, and social stories, that can help children with ASD develop their cognitive, social, and communication skills. The app is based on scientific research and has been tested with a population in an Autism center. The app is designed to be user-friendly and accessible to children with ASD. This uses an architectural solution that enables the interoperability between an autistic person and caregivers to contribute on the treatment and monitoring to persons with ASD. The solution is implemented by creating a mobile application and tested with a population in an Autism center. The app made a positive treatment contribution to a person with ASD.

In the year 2020, the paper [9] describes a mobile application which is done for ASD kids who are in 1-8 age. This application is 'AutiSpark'. AutiSpark is an educational program made specifically for children with Autism Spectrum Disorder (ASD) with learning games tested and approved by special educators and occupational therapists. Through their pictures, children can learn to associate objects, understand emotions, classify, recognize sounds and words, and much more. More features include letters, letter combinations and word recognition, increasing basic math skills to recognize numbers and patterns, social videos, tracking games, memory games and puzzles, etc.

In the year 2019, the paper [10] proposes a new multi-agent architecture of an emotional intelligent e-learning system, it aims to help ASD children overcome learning impairments. The paper describes the roles and functions of each agent, such as the learner agent, the tutor agent, the emotion agent, the pedagogical agent, and the interface agent. The paper also presents the communication protocols and the algorithms used to model the learner's profile, to select the appropriate content, and to evaluate the learner's performance and satisfaction. Some of the main features, algorithms, and architectures of this paper are: The system uses a multi-agent architecture, which is a distributed system composed of autonomous and cooperative entities that can communicate and interact with each other. The system uses an emotional model, which is a representation of the learner's emotional state based on facial expressions, voice, and physiological signals. The system uses a learner model, which is a representation of the learner's cognitive and affective characteristics, such as learning style, preferences, goals, knowledge, and emotions. The system uses a content model, which is a representation of the learning material, such as topics, concepts, activities, and media. The system uses a pedagogical model, which is a representation of the teaching strategies, such as feedback, guidance, adaptation, and motivation. The system uses a Bayesian network, which is a probabilistic graphical model that can infer the learner 'profile and the optimal content based on the evidence and the prior knowledge. The system uses a fuzzy logic, which is a mathematical framework that can deal with uncertainty and vagueness in the decision making process. The system uses a genetic algorithm, which is a bio-inspired optimization technique that can find the best solution among a population of possible solutions.

The paper [11] describes application for children with ASD that uses gamification and animation techniques to enhance their learning experience. The paper explains the design and development process of the application, which includes the following steps: analysis, design, development, implementation, and evaluation. The paper also discusses the algorithms used to generate the questions, to provide feedback, and to track the progress of the learners. The application uses a gamification technique, which is the use of game elements and mechanics in non-game contexts to increase engagement and motivation. Animation technique use of moving images and sounds to create an interactive and immersive environment. This application uses a randomization algorithm, which is an algorithm that generates random numbers or sequences to create different questions and scenarios for each learner. The application uses a scoring algorithm, which is an algorithm that assigns points or grades to the learner's answers based on the correctness and difficulty level. The application uses a tracking algorithm, which is an algorithm that records and analyzes the learner's performance and behavior over time.

When compared to the past implementations discussed above, "SIPNENA" has existing functionalities and new features in each component. Some innovative features have been added such as giving the child the basic knowledge of mathematics and English at an early stage and making arrangements to maintain a conversation with the help of a chatbot to develop communication skills. User-Centered Interaction Design has been used throughout the entire implementation to make the application more convenient to use..

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III. Methodology

The problem's solution has broken down into five sections. Each section has its solution to the challenges that the survey addresses.

A. *Developing an Augmentative and Alternative Communication System*

1) *System implementation*

The Augmentative and Alternative Communication (AAC) system facilitates communication development by integrating basic vocabulary images and words. The core board encompasses the most frequently used vocabulary, such as "you," "I," "get," "do," and "where," which constitute a significant portion of everyday communication. The visual representation of symbols is consistent with the user's high-tech device, fostering ease of learning and generalization. Furthermore, the grid size is tailored to accommodate the user's requirements, ensuring an optimal and user-friendly experience. The color scheme employed in the creation of this AAC system adheres to established standards, enhancing clarity and familiarity for the users.



Fig.1. Workflow of the AAC system implementation.

The color selection for the AAC system is a crucial consideration. Instead of arbitrary choices, standardized colors were employed, utilizing blue for adjectives, green for verbs, yellow for pronouns, orange for nouns, white for conjunctions, pink for prepositions and social words, purple for questions, brown for adverbs, red for essential function words, negation, and emergency words, and gray for determiners.[12]

Data collection

A speaking-assistance system has been developed to enhance the communication abilities of children with ASD, even at the young age of six years old. The system utilizes pre-trained, concise, and straightforward conversational questions to engage with users. Data was collected through a combination of online and physical methods, with the primary objective of identifying the set of questions required for training the chatbot. The chatbot training involved approximately 1000 questions, with 500 questions obtained from 50 individuals through physical interactions and another 500 questions from 50 individuals online. The data was gathered from non-ASD children above the age of 18.

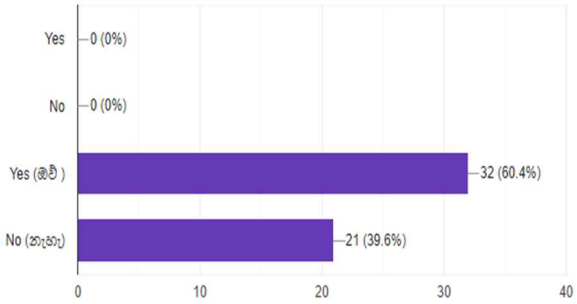


Fig. 2. Data collection process for training the conversational chatbot.

The study revealed that over 20% of children struggle with expressing their opinions as shown in Fig. 2. To address this challenge, a custom neural network was developed using a Keras sequential model. The model was optimized using stochastic gradient descent (SGD), with classifier cross-entropy as the loss function and accuracy as the training measure. Natural Language Processing (NLP) techniques were employed, including pre-processing input sentences, encoding them into individual words, and lemmatizing them using the Natural Language Toolkit for working with human language data.

B. *Activity monitoring using facial imotions*

The focus of this component is to track the engagement of autistic children with on-screen learning activities. While the user interacts with educational activities on a tablet, the device's front camera is utilized to monitor their involvement. The study employs computer vision and deep learning techniques to monitor children with ASD using facial emotions. A ResNet50 pre-trained Convolutional Neural Network (CNN) model is used for face recognition, capturing real-time video frames at specific time intervals. The model is trained to predict real-time emotions using a dataset of 11,500 sample images and tested on 2,500 images from six classes: angry, disgusted, happy, neutral, sad, and surprised.

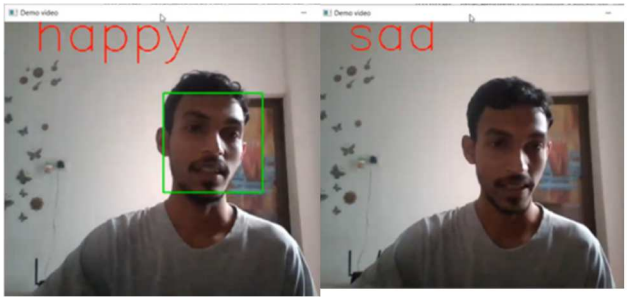


Fig. 3. Real-time emotion detection using facial expressions.

The system continuously monitors a child's emotions during on-screen activities, generating reports detailing the dominant emotions and important details for analyzing children with ASD. The conceptual diagram is shown in Fig. 5. Based on the child's past dominant emotions, the system suggests the next preferred activity. These reports are stored in a database for

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reference by parents and teachers. If a child's engagement with an activity exceeds a predetermined threshold percentage, an alert is sent to the parent's account.

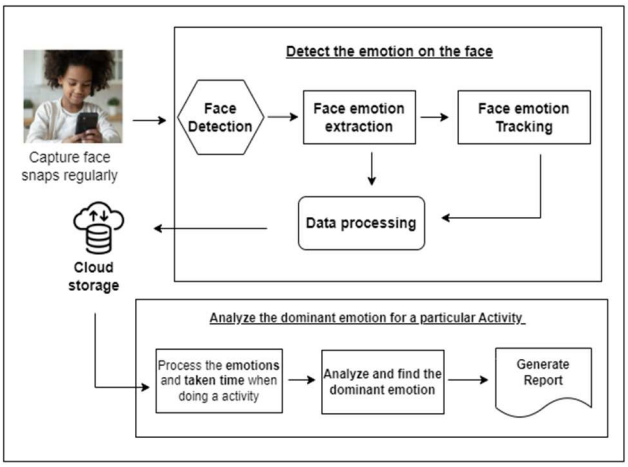


Fig. 4. Emotion detection model architecture

C. Develop Customize English language support system

In the development of an interactive mobile application tailored for children with autism, the primary goal is to facilitate the learning of the English alphabet and common words through an innovative approach that combines technology with pedagogical strategies. This endeavor encompasses a multifaceted methodology, structured around four pivotal stages: initial alphabet instruction, comprehension testing, the integration of common words with gamified elements, and continuous progress monitoring. The foundation of this educational tool is bolstered by a rich collection of image datasets, comprising over 200 sets, which cater to a spectrum of lighting conditions for the letters. Furthermore, the incorporation of a Convolutional Neural Network (CNN) model serves to enhance the effectiveness of spoken language analysis, thereby enabling real-time feedback, personalized learning experiences, and an engaging interface that caters specifically to the needs of children with autism.

Teaching the Alphabet

The application will teach children to write English alphabet letters on a dotted line, presenting each letter in context to aid comprehension. The application will provide immediate feedback on the child's writing, awarding points for correct responses and deducting attempts for incorrect ones, ensuring a comprehensive learning experience.

Testing Comprehension

Following the completion of alphabet lessons, the application incorporates a short quiz to evaluate the child's grasp and retention of the material. This formative assessment serves as a diagnostic tool, allowing for the adjustment of future lessons based on the child's performance. It provides valuable insights into the child's strengths and areas needing improvement, thus tailoring the learning path to meet individual needs effectively.

Teaching common words and implementing gamification elements

The application introduces everyday words using alphabet lessons, requiring children to write them on a dotted line. Successful completion earns points, while unsuccessful attempts result in no points. To increase engagement, the application uses gamification elements like leaderboards, badges, and rewards, encouraging competition and a sense of achievement upon reaching certain milestones.

Assessing progress

This research focuses on developing a Convolutional Neural Network (CNN) model to support interactive English learning for children with autism. The model is trained on a dataset of audio samples and preprocessed into a Mel spectrogram for consistent analysis. The CNN architecture, built with TensorFlow and Keras, consists of convolutional and pooling layers for feature extraction and dense layers for classification. The training process employs the Adam optimizer and categorical cross-entropy loss function, with early stopping to avoid overfitting. The model's performance is evaluated using a test set, and once satisfactory, it is deployed within a mobile application. This application offers real-time analysis of the child's spoken input, providing immediate feedback on pronunciation and comprehension, and fostering an engaging and personalized learning environment that enhances language acquisition for children with autism.

D. Develop a numeracy skill improvement system

In this study, a novel approach is implemented to facilitate counting skills in autistic children, focusing on the numerical range from zero to nine. The intervention involves presenting numbers to the participants and assessing their ability to make accurate judgments. Points are awarded for correct pronunciation, while instances of incorrect pronunciation are utilized as opportunities for improvement and reinforcement.

Data collection

In this research endeavor, a cohort comprising 1000 students, specifically enrolled in 2nd and 3rd grades, actively participated in the process of articulating numbers spanning from zero to the name of an advanced mathematical formula. The meticulous execution of this auditory task was facilitated using Audacity software, a sophisticated audio editing tool. Within this digital environment, extraneous sounds were methodically eradicated, ensuring the isolation and preservation of the numerical pronunciations. The process yielded a comprehensive dataset, a symphony of voices from 1000 children, each contributing their unique articulation of one-digit numbers. This extensive collection resulted in the curation of 10,000 individual sound samples, forming the bedrock of our investigation into the nuances of numerical pronunciation proficiency among students in the early grades. This methodology, blending technology with pedagogy, provides a robust foundation for analyzing and understanding the intricacies of verbal numerical expression in a diverse educational setting.

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E. *Providing an optimal end-user experience by involving users*

The research focuses on UI/UX engineering technologies for the user journey of a mobile application. Design thinking is an approach that uses design methodologies to re-evaluate paradigms and modify existing technology to create new procedures suitable for users. User-Centered Design (UCD) is chosen as the best solution to ensure clear and uncomplicated designs. The research uses a pragmatic strategy that includes qualitative and quantitative methods to gain a deep understanding of users. The main goal is to demonstrate high-fidelity prototypes of an e-learning system on portable devices that are easy to use, have a minimal learning curve, and do not require excessive time to understand, implement, and satisfy users' expectations. The research objectives include identifying the most appropriate data collection and analysis methods, real-world problems, and interaction design concepts. The survey-based technique was chosen for data collection and analysis, simplifying participant responses. Potential problems were identified, and changes to the mobile application were made using interactive design principles.

Overall system design was finalized and implemented after making the modifications for the interfaces by taking all the responses of the participants and the UCID methodology into account.

IV. DISCUSION

The proposed research endeavor offers a comprehensive solution to address the challenges faced by the target user group. The user will be required to register for a system account to access the full range of features and functionalities.

SIPNENA is a mobile application that performs real-time emotion detection while children with Autism Spectrum Disorder (ASD) engage in learning activities. Research efforts have been directed toward emotion recognition in children with ASD, with the aim of providing them with a deeper understanding of emotions to enhance their recognition abilities. The application employs six distinct emotional categories: angry, disgusted, happy, neutral, sad, and surprised, tailored to the children's emotional states [13].

SIPNENA is a comprehensive platform that provides an Augmentative and Alternative Communication (AAC) system for communication and speech training, with the primary objective of improving the speaking skills of young children. Unlike most AAC systems, which are typically emulated on phones or tablet devices [14], SIPNENA has developed a Sinhala version of the core board. This unique feature enables children to communicate using their voice, a departure from traditional systems that rely solely on typing [15]. SIPNENA's approach transcends conventional AAC systems, allowing children to effectively communicate through their voices, fostering a more natural and engaging experience.

While numerous applications have been developed to enhance the communication skills of children with ASD, there is a paucity of e-learning applications designed to expand their knowledge base. Some applications focus solely on improving children's numeracy skills [16], while others are exclusively dedicated to enhancing English language proficiency. However, SIPNENA has adopted a holistic approach, incorporating various activities to foster the development of multiple skills, including communication, English language, and mathematics.

SIPNENA is a user-friendly application tailored for young children, with a strong emphasis on functionality and an intuitive user interface. It offers a comprehensive solution by amalgamating features such as emotion detection, communication skill development, and subject-related activities. Unlike other applications that provide these features in isolation, SIPNENA consolidates them into a unified platform, facilitating convenient access and utilization for children. This integrated approach distinguishes SIPNENA as a unique and valuable tool for children with ASD.

V. CONCLUSTION

This article presents SIPNENA, an innovative e-learning and communication platform designed for children with Autism Spectrum Disorder (ASD). It integrates features like a chatbot, emotion recognition, an AAC keyboard, and activities for English and Math learning. Future work aims to expand language support, broaden subject-related activities, and enhance user experience and accessibility. The goal is to provide a comprehensive, inclusive, and engaging learning environment that empowers children with ASD to develop essential skills, while contributing to the field of assistive technology and inclusive education practices.

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